

2026 AHA/ACC/HRS PRACTICE GUIDELINE FOR ATRIAL FIBRILLATION

Clinical Evaluation, Diagnosis & Classification Framework

1. CLINICAL OVERVIEW & EPIDEMIOLOGY

Atrial fibrillation (AF) is the most prevalent sustained supraventricular arrhythmia globally. AF is associated with a 5-fold increase in thromboembolic stroke, a 3-fold elevation in heart failure risk, and a 1.5- to 1.9-fold increase in all-cause mortality. Early identification and systematic stroke risk stratification are vital components of modern cardiovascular care.

2. DIAGNOSTIC CRITERIA & WORKUP

A definitive clinical diagnosis of AF requires documentation on a standard 12-lead electrocardiogram (ECG) or a continuous rhythm strip of at least 30 seconds showing irregular R-R intervals without discernible, organized P waves. Initial comprehensive evaluation must include:

- ? Transthoracic echocardiography (TTE) to evaluate left atrial size, LVEF, and valvular morphology.
- ? Serum thyroid-stimulating hormone (TSH) to exclude hyperthyroidism.
- ? Complete metabolic panel and serum creatinine/eGFR for baseline renal assessment.
- ? Complete blood count (CBC) to screen for occult anemia.

3. ARRHYTHMIA CLASSIFICATION PATTERNS

- ? Paroxysmal AF: Spontaneously terminates or converts with intervention within 7 days of onset.
- ? Persistent AF: Continuous AF sustained beyond 7 days, requiring pharmacological or electrical cardioversion.
- ? Long-Standing Persistent AF: Continuous AF lasting longer than 12 months in duration.
- ? Permanent AF: Patient and clinician agree to cease further rhythm control strategies.

4. INITIAL STABILIZATION

Patients presenting with acute hemodynamically unstable AF (hypotension, acute pulmonary edema, angina) mandate immediate synchronized electrical cardioversion regardless of arrhythmia duration.

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Thromboembolic Stroke Stratification & Bleeding Risk Assessment

5. STROKE RISK STRATIFICATION (CHA2DS2-VASc)

The CHA2DS2-VASc scoring model is the gold standard for thromboembolic risk determination in non-valvular AF:

- ? Congestive Heart Failure or LVEF $\leq$ 40% (+1 point)
- ? Hypertension / resting BP $\geq$ 140/90 mmHg (+1 point)
- ? Age $\geq$ 75 years (+2 points)
- ? Diabetes Mellitus (+1 point)
- ? Prior Stroke, TIA, or Thromboembolism (+2 points)
- ? Vascular Disease (Prior MI, PAD, aortic plaque) (+1 point)
- ? Age 65 to 74 years (+1 point)
- ? Sex Category Female (+1 point)

RECOMMENDATIONS FOR ANTICOAGULATION:

- ? Score $\geq$ 2 in Men or $\geq$ 3 in Women: Oral anticoagulation is strongly recommended (Class I, Level A).
- ? Score = 1 in Men or = 2 in Women: Oral anticoagulation should be considered based on clinical shared decision-making.
- ? Score = 0 in Men or = 1 in Women: No antithrombotic therapy is recommended.

6. BLEEDING RISK EVALUATION (HAS-BLED)

Evaluating bleeding risk helps identify modifiable risk factors rather than serving as a reason to withhold anticoagulation.

A HAS-BLED score $\geq$ 3 indicates high bleeding risk requiring frequent clinical monitoring.

MODIFIABLE BLEEDING RISK FACTORS:

- ? Uncontrolled hypertension (systolic BP $>$ 160 mmHg).
- ? Concomitant use of antiplatelet agents (aspirin, clopidogrel) or NSAIDs.
- ? Excessive alcohol intake ($>$ 8 units per week).
- ? Labile INR values in patients on Warfarin.
- ? Modifiable liver or renal impairment.

TABLE 1: CHA2DS2-VASc SCORE VS ANNUAL STROKE RISK

Score 0:	0.2% annual stroke risk (No therapy)
Score 1:	0.6% annual stroke risk (Consider DOAC)
Score 2:	2.2% annual stroke risk (DOAC Indicated)
Score 3:	3.2% annual stroke risk (DOAC Indicated)
Score 4+:	4.8% - 11.2% annual stroke risk (High Risk)

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Direct Oral Anticoagulants (DOACs): Pharmacotherapy & Renal Adjustments

7. FIRST-LINE ORAL ANTICOAGULATION (DOACs)

Direct Oral Anticoagulants (DOACs) are recommended in preference to vitamin K antagonists (Warfarin) for stroke prevention in non-valvular AF due to superior safety profiles and significantly reduced rates of fatal intracranial hemorrhage.

RECOMMENDED DOAC REGIMENS:

? Apixaban: Standard dose is 5 mg orally twice daily.

Dose reduction to 2.5 mg orally twice daily is required if the patient meets at least TWO of the following criteria: Age $\geq$ 80 years, Body weight $\leq$ 60 kg, or Serum creatinine $\geq$ 1.5 mg/dL (133 μ mol/L).

? Rivaroxaban: Standard dose is 20 mg orally once daily taken with the evening meal (food is required to achieve consistent oral bioavailability).

Dose reduction to 15 mg orally once daily is indicated for patients with moderate renal impairment (Creatinine Clearance CrCl 15 to 49 mL/min).

? Dabigatran: Direct thrombin inhibitor. Standard dose is 150 mg orally twice daily.

Dose reduction to 75 mg orally twice daily is indicated for CrCl 15 to 30 mL/min.

? Edoxaban: Factor Xa inhibitor. Standard dose is 60 mg orally once daily.

Reduce to 30 mg orally once daily if CrCl 15 to 50 mL/min or body weight $\leq$ 60 kg.

CONTRAINDICATIONS TO DOACs:

DOACs are strictly contraindicated in patients with mechanical prosthetic heart valves or moderate-to-severe mitral stenosis. Warfarin (target INR 2.0 to 3.0) remains the mandatory anticoagulation choice for mechanical valves.

TABLE 2: DOAC FIRST-LINE DOSING MATRIX

Apixaban: 5 mg bid (2.5 mg bid if 2 of: age $\geq$ 80, wt $\leq$ 60kg, Cr $\geq$ 1.5)

Rivaroxaban: 20 mg qd with food (15 mg qd if CrCl 15-49 mL/min)

Dabigatran: 150 mg bid (75 mg bid if CrCl 15-30 mL/min)

Edoxaban: 60 mg qd (30 mg qd if CrCl 15-50 mL/min)

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Rate vs Rhythm Control Strategies & Catheter Ablation Protocols

8. RATE CONTROL PHARMACOTHERAPY

Ventricular rate control is initial therapy for the majority of symptomatic AF patients. Target resting heart rate is < 100 to 110 beats per minute.

FIRST-LINE RATE CONTROL AGENTS:

? Beta-Blockers (Preserved or Reduced LVEF):

- Metoprolol Succinate: 50 to 200 mg orally once daily.
- Bisoprolol: 2.5 to 10 mg orally once daily.
- Carvedilol: 3.125 to 25 mg orally twice daily.

? Non-Dihydropyridine Calcium Channel Blockers

(Preserved LVEF > 40% only):

- Diltiazem Extended Release: 120 to 360 mg orally once daily.

- Verapamil: 120 to 360 mg orally once daily.

CAUTION: CCBs are contraindicated in heart failure with reduced ejection fraction (HFrEF).

? Digoxin: 0.125 to 0.25 mg orally once daily. Useful in sedentary patients or as add-on in HFrEF.

9. RHYTHM CONTROL & CARDIOVERSION PROTOCOLS

Early rhythm control improves cardiovascular outcomes in patients diagnosed within 1 year.

ANTICOAGULATION PERI-CARDIOVERSION MANDATE:

For AF of ≥ 48 hours duration or unknown duration, therapeutic anticoagulation (DOAC or Warfarin) is mandatory for at least 3 consecutive weeks prior to elective cardioversion, and MUST be continued for a minimum of 4 weeks post-cardioversion regardless of successful sinus rhythm restoration.

10. CATHETER ABLATION INDICATIONS

Pulmonary vein isolation (PVI) catheter ablation is recommended as Class I therapy for symptomatic paroxysmal AF refractory or intolerant to at least one antiarrhythmic drug (AAD), and as first-line therapy in patients with heart failure and reduced ejection fraction to improve survival.